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Laxative Effects of *Rosa damascena* Mill in Dogs

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Abstract

Abbaszadeh, M., Kazerani, H.R. and Kamrani, A. 2010. Laxative effects of *Rosa damascena* mill in dogs. J. Appl. Anim. Res., 38: 89-92.

The laxative/purgative effects of Rosa damascena was studied in dogs. Five Groups of animals (n=5) received the boiled extract of Rosa damascena at doses 0.5-8 times that traditionally suggested for human use (180mg/kg) daily. The negative and the positive groups received placebo or lactulose respectively. During 10 d of the experiment, fecal water content was measured and the dogs were examined for the incidence of diarrhea. In addition, all dogs were monitored for their weight, ECG, temperature and their respiration and heart rates. Rosa damascena caused diarrhea in a dose dependent manner. The fecal water content, however, did not change significantly. The animals receiving the highest dose of the extract showed sedation and a slight weight loss. However, no other side effects were observed in treated animals. It is concluded that boiled extract of Rosa damascena may be of therapeutic value as laxative in dogs.

Keywords: *Rosa damascena*, laxative, herbal medicine.

Introduction

Rosa damascena Mill has been traditionally used in treatment of constipation as herbal medicine in human being without apparent harmful effects. Several other therapeutic effects including antitussive (Shafei *et al.*,

2003), anti-nociceptive (Rakhshandeh *et al.*, 2008), hypoglycemic (Gholamhoseinian, 2009) and bronchodilator (Boskabady *et al.*, 2006) properties have been described for the flowers of *Rosa damascena* Mill in recent studies.

The boiled extract of *Rosa damascena* Mill has been traditionally used for treatment of constipation in human (Zargari, 1992). To the best of our knowledge, however, there is no scientific report regarding its laxative effects

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in human or in animals. The main target of the present study was to preliminarily evaluate the possible use of this plant as laxative in dogs.

Materials and Methods

The basal dosage (N) of the plant was extrapolated according to its traditional usage in human as 180 mg/kg. Dried flower petals of *Rosa damascena* were boiled in distilled water (120 g/l) for 10 min. The filtrate was collected and stored at 4°C for a maximum period of 3 d before being fed to the animals. Each gram of the original flower petal was calculated to yield 0.52 g dry matter in the extract. The undiluted extract was used for the maximum dosage (8N). The extract was serially diluted with distilled water to yield other doses of 4N, 2N, N and 1/2N.

Thirty three healthy dogs (20.1±1.5kg) were randomly assigned to 7 experimental groups. Five experimental groups (n=5) received varying doses of the herb extract (1/2N, N, 2N, 4N and 8N: 90-1440 mg/kg) daily. The 6th group (n=4) received a similar volume (12 ml/kg) of placebo (distilled water) and was considered as the negative control. The 7th

group (n=4) received lactulose (300 mg/kg/d) and served as the positive control.

The animals in different experimental groups received either the extract, lactulose or the placebo for 10 successive days. The respiration and the heart rates, temperature and ECG were recorded at days 0, 1, 3, 7 and 10. In addition to visual monitoring for incidence of diarrhoea (defined as lack of consistency), fecal water content was measured on rectal samples at mentioned dates according to the method used by Saito *et al.*, 2000).

Statistical analysis and drawing of the figures were performed using GraphPad Prism v4.0 (GraphPad Software, USA). Statistical comparisons were performed using two-way analysis of variance (ANOVA) followed by Bonferroni post tests. However, one-way ANOVA followed by Dunnett's test were used for weight gain.

Results and Discussion

Rectal temperature, respiration and heart rates, ECG of all the dogs on all the days were within normal physiological limits and did not differ among groups/ days ($P>0.05$). The boiled

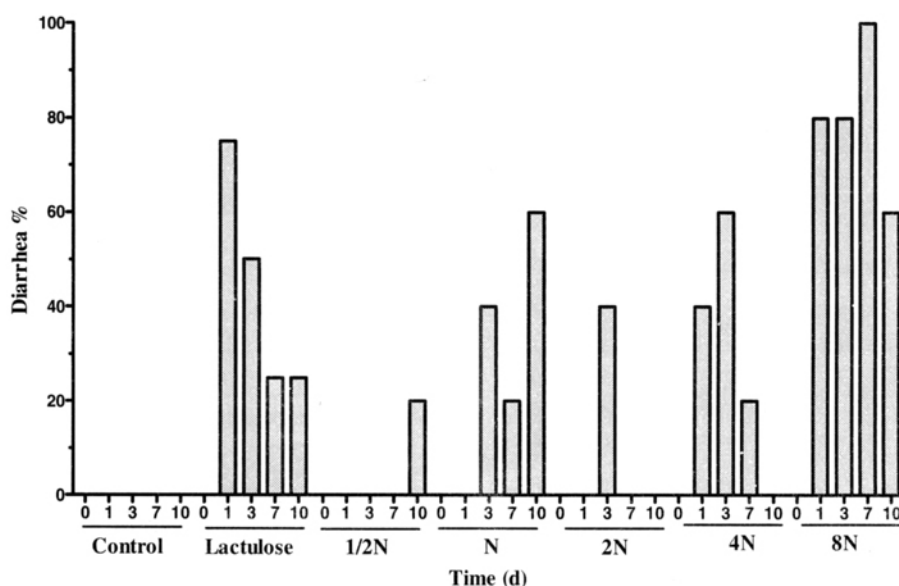


Fig. 1. The percentage of diarrhea incidence in different experimental groups. The control group received placebo, the second group received lactulose (300 mg/kg/d) and the test groups received exponential doses of *Rosa damascena* (N=180 mg/kg/d) during 10 d of the experiment.

Table 1

The percentage of fecal water content in different experimental groups. The dogs received placebo (control), lactulose (300 mg/kg/d) or exponential doses of *Rosa damascena* (N=180 mg/kg/d) during 10 d of the experiment

	Days	0	1	3	7	10
Control	Mean	78	78.3	79.1	75.6	67.5
	SEM	6.4	4.6	2.8	4.3	7.2
Lactulose	Mean	82.5	65.3	73.4	76.4	82.4
	SEM	2.1	0	2.8	1.6	5.1
1/2 N	Mean	82	62.5*	79.2	80	72.4
	SEM	1	3.6	2.1	0.9	0.1
N	Mean	76.8	66.6	80.9	75.8	73.2
	SEM	7.2	2.5	2.9	2.4	2.3
2N	Mean	83	69.1	79.5	72.7	77.4
	SEM	2.1	3.2	3.9	2.5	5.6
4N	Mean	75.8	69.5	75.8	73.8	71.2
	SEM	2.1	11.1	10.5	8.4	2.3
8N	Mean	82.2	80.5	77.6	-	85.1
	SEM	3.1	0	0	-	2.8

extract of *Rosa damascena* caused efficient laxative/purgative effects in dogs (Fig.1). The effect was dose dependent and the extract caused diarrhea even with the lowest dose used (90 mg/kg/d).

The animals seemed to show tolerance to higher doses of *Rosa damascena*. This usually happens with other laxative/purgative drugs and was also seen with lactulose in this study. The time needed to develop tolerance against *Rosa damascena*, however, seemed to be longer compared to lactulose.

In this study, we failed to show significant changes in fecal water content due to either lactulose or *Rosa damascene* (Table 1). The percentage of fecal water content in this study was similar to those previously reported (Ueno *et al*, 2002). Consistent to our study, the latter study showed that minimal changes in fecal water content could profoundly affect fecal consistency. Therefore, the lack of statistical significance may be due to the rather small difference in fecal water in dogs during diarrhea.

Rosa damascena did not cause any apparent side effects even with the highest dose during 10 d of the experiment. The respiration and the heart rates, as well as the

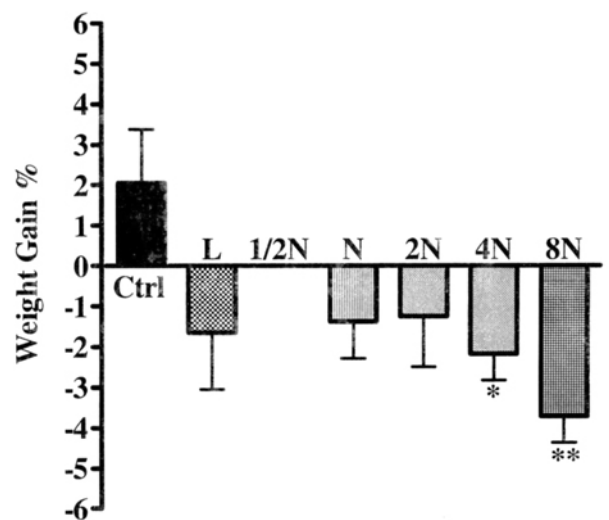


Fig. 2. The percentage of weight gain in different experimental groups during the experiment. Ctrl: negative control, L: lactulose (300 mg/kg/d), N: boiled extract of *Rosa damascena* (180 mg/kg/d). *P<0.05, **P<0.01.

temperature, were within the normal ranges in all experimental groups. Visual examination of the animals revealed no side effects in treated animals. The dogs receiving the highest dose of the extract, however, showed symptoms of sedation and anorexia especially following d 7 of the experiment.

Rosa damascena caused a slight dose-dependent weight loss in treated dogs (Fig. 2). A slight weight loss was also observed in dogs receiving lactulose. This might be due to diarrhea induced malabsorption or dehydration (DiBartola, 2000). As mentioned above, the highest dose of *Rosa damascena* caused anorexia. In addition, our recent studies (not published) suggest that this herb might delay gastric emptying in rats. However, the exact mechanism behind weight loss demands further investigation.

In conclusion, *Rosa damascena* may be successfully prescribed as laxative/purgative in dogs.

Acknowledgements

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- एम. अब्बसजादेह, एच.आर. काजेरानी, ए. कामरानी। कुत्तों में रोसा डैमेसेना का मृदु विरेचक प्रभाव।
- कुत्तों में रोसा डैमेसेना के मृदु विरेचक/विरेचक प्रभाव का अध्ययन किया गया। रोसा डैमेसेना को उबाल कर बनाए गये निष्कर्ष को मनुष्यों में दी जाने वाली मात्रा (180 मिग्रा/किग्रा शरीर भार) का 0.5 से 8 गुना तक कुत्तों को पांच वर्गों में दिया गया। ऋणात्मक और धनात्मक कंट्रोल में क्रमशः कूट-भेषज या लैक्टुलोज दिया गया। परीक्षण के 10 दिनों में मल में जल की मात्रा पायी गयी और अतिसार के लिए कुत्तों की जांच की गयी। इस के साथ ही सभी कुत्तों के भार, ईसीजी, तापक्रम, श्वसन गति और हृदय गति की भी निगरानी की गयी। रोसा डैमेसेना की मात्राओं के अनुसार अतिसार के लक्षण प्रगट हुये। वैसे, मलीय जल की मात्रा दिए गये कुत्तों में शामक प्रभाव और शरीर भार में थोड़ी कमी पायी गयी। वैसे, उपचारित पशुओं में कोई अन्य अतिरिक्त प्रभाव नहीं पाया गया। निष्कर्ष यह निकला कि कुत्तों के लिए रोसा डैमेसेना के काढ़े का मृदु विरेचन हेतु औषधीय मूल्य है।